TM Modes in Rectangular Waveguide

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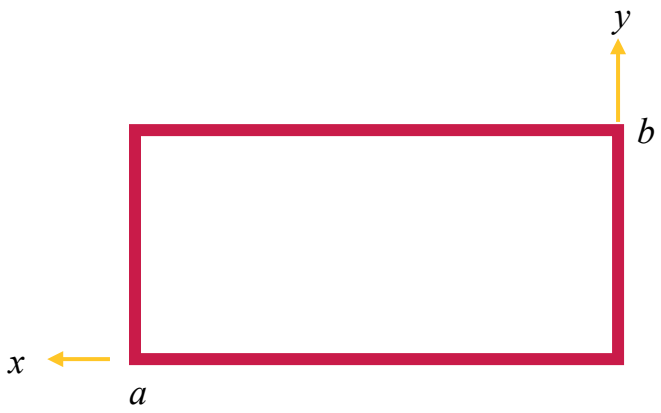
Lecture Objective

By the end of this lecture, you will be prepared to:

- Evaluate the characteristics of TM modes in rectangular waveguide.

Rectangular Waveguide Geometry



- Consider a hollow tube waveguide comprising a rectangular cross-section.
- By convention, the longer side of the rectangle is placed along the x-axis and the value of its length is designated as a; the value of the length of the shorter side is designated as b.

Modes Supported by RWG

- As a single conductor waveguide, the structure cannot support a TEM mode.
- Any mode supported by a homogeneously filled RWG will be either TE or TM.
- As we shall see, the dominant mode is TE_{10}.

TM Guided Wave Eigenmodes

- Eigenmodes are solutions to:

$$\left(\nabla_t^{\,2} + k_c^{\,2}\right)\tilde{e}_z = 0 \quad + BCs$$

(1)

$$\tilde{e}_x = -\frac{1}{k_c^{\,2}}\left(\gamma \frac{\partial \tilde{e}_z}{\partial x}\right)$$

$$\tilde{e}_y = -\frac{1}{k_c^{\,2}}\left(\gamma \frac{\partial \tilde{e}_z}{\partial y}\right)$$

$$\tilde{h}_x = -\frac{1}{k_c^{\,2}}\left(j\omega\varepsilon \frac{\partial \tilde{e}_z}{\partial y}\right)$$

$$\tilde{h}_y = -\frac{1}{k_c^{\,2}}\left(j\omega\varepsilon \frac{\partial \tilde{e}_z}{\partial x}\right)$$

(2)

EVP for TM Modes

- DE:

$$\frac{\partial^2 \tilde{e}_z(x,y)}{\partial x^2} + \frac{\partial^2 \tilde{e}_z(x,y)}{\partial y^2} + k_c{}^2 \tilde{e}_z(x,y) = 0$$

- BC:

$$\tilde{e}_z(x=0,y) = \tilde{e}_z(x=a,y) = 0$$

$$\tilde{e}_z(x,y=0) = \tilde{e}_z(x,y=b) = 0$$

Solution Technique: Separation of Variables

- Assume:

$$\tilde{e}_z(x,y) = X(x)\cdot Y(y)$$

- Substitution of above into the DE yields:

$$Y(y)\frac{d^2X(x)}{dx^2} + X(x)\frac{d^2Y(y)}{dy^2} + k_c{}^2 X(x)Y(y) = 0$$

2. Solution Technique: Separation of Variables

- Division of each term by $X(x)Y(y)$ yields:

$$\frac{1}{X(x)}\frac{d^2X(x)}{dx^2}+\frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2}+k_c{}^2=0$$

- For above equation to hold for every value of x and every value of y, we must have

$$\frac{1}{X(x)}\frac{d^2X(x)}{dx^2}=-k_x{}^2 \qquad \frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2}=-k_y{}^2$$

3. Solution Technique: Separation of Variables

- Separated DEs:

$$\frac{d^2X(x)}{dx^2} + k_x{}^2 X(x) = 0$$

$$\frac{d^2Y(y)}{dy^2} + k_y{}^2 Y(y) = 0$$

- Constraint equation:

$$k_x{}^2 + k_y{}^2 = k_c{}^2$$

- Separated BCs:

$$X(0) = X(a) = 0$$

$$Y(0) = Y(b) = 0$$

4. Solution Technique: Separation of Variables

- Separated EVPs:

$$\frac{d^2X(x)}{dx^2} + k_x{}^2 X(x) = 0 \qquad \frac{d^2Y(y)}{dy^2} + k_y{}^2 Y(y) = 0$$

$$X(0) = X(a) = 0 \qquad Y(0) = Y(b) = 0$$

5. Solution Technique: Separation of Variables

- Solutions to separated EVPs:

$$X(x) = \sin\left(\frac{m\pi}{a}x\right) \qquad Y(y) = \sin\left(\frac{n\pi}{b}y\right)$$

$$k_x = \frac{m\pi}{a}, \quad m = 1, 2, 3, \cdots \qquad k_y = \frac{n\pi}{b}, \quad n = 1, 2, 3, \cdots$$

Solution to EVP for TM Modes

- Solution to original EVP:

$$\tilde{e}_z(x,y) = \sin\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)$$

$$k_c = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}, \quad m = 1,2,3,\cdots; n = 1,2,3,\cdots$$

Propagation Constant and Cutoff Frequencies for TM Modes

- Propagation constant:

$$\gamma_{mn} = \sqrt{k_c^{\,2} - k^2} = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega^2 \mu\varepsilon}$$

- Cutoff frequency:

$$k_c = \omega_c \sqrt{\mu\varepsilon} = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

$$f_{mn} = \frac{c}{2\sqrt{\mu_r \varepsilon_r}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

Transverse Fields for TM Modes

- Transverse fields:

$$\tilde{e}_x(x,y) = -\frac{\gamma_{mn}}{k_c^{\,2}}\frac{\partial \tilde{e}_z}{\partial x} = -\frac{\gamma_{mn}}{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}\left(\frac{m\pi}{a}\right)\cos\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)$$

$$\tilde{e}_y(x,y) = -\frac{\gamma_{mn}}{k_c^{\,2}}\frac{\partial \tilde{e}_z}{\partial y} = -\frac{\gamma_{mn}}{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}\left(\frac{n\pi}{b}\right)\sin\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)$$

2. Transverse Fields for TM Modes

- Transverse fields:

$$\tilde{h}_x(x,y) = \frac{j\omega\varepsilon}{k_c^{\,2}}\frac{\partial \tilde{e}_z}{\partial y} = \frac{j\omega\varepsilon}{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}\left(\frac{n\pi}{b}\right)\sin\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)$$

$$\tilde{h}_y(x,y) = -\frac{j\omega\varepsilon}{k_c^{\,2}}\frac{\partial \tilde{e}_z}{\partial x} = -\frac{j\omega\varepsilon}{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}\left(\frac{m\pi}{a}\right)\cos\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)$$

Wave Impedance for TM Modes

- Wave impedance:

$$\gamma_{mn} = jk\sqrt{1-\left(\frac{f_{mn}}{f}\right)^2} = j\omega\sqrt{\varepsilon\mu}\sqrt{1-\left(\frac{f_{mn}}{f}\right)^2}$$

$$Z^{TE} = \frac{\tilde{E}_x}{\tilde{H}_y} = \frac{-\tilde{E}_y}{\tilde{H}_x} = \frac{\gamma_{mn}}{j\omega\mu} = \eta\sqrt{1-\left(\frac{f_{mn}}{f}\right)^2}$$

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